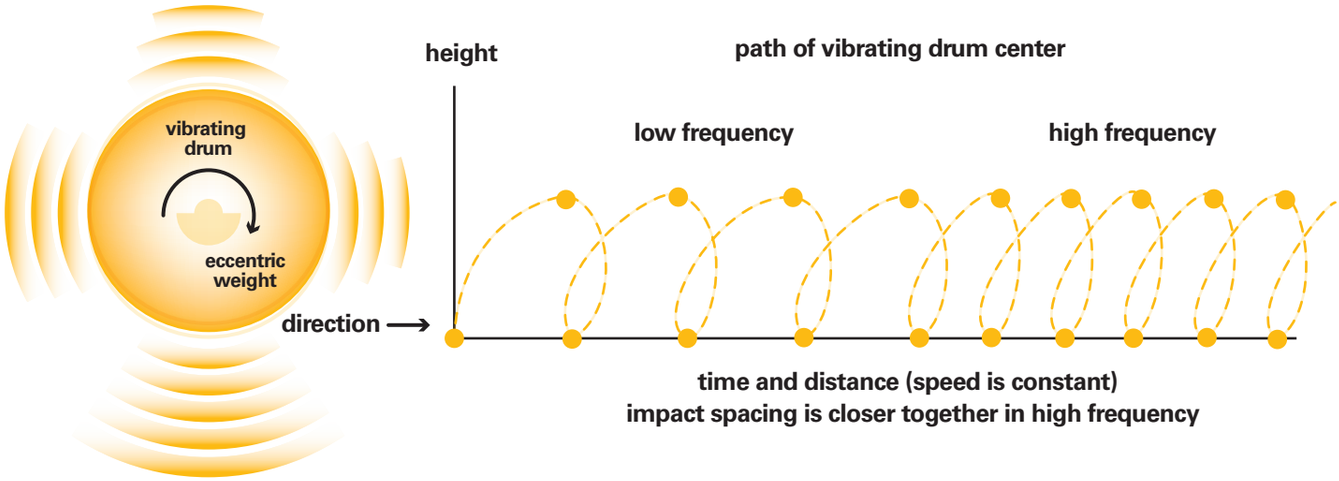
FREQUENCY



[VIBRATION EQUALS FREQUENCY]

Vibratory force is referred to as frequency. Frequency is defined as the number of times that the drum hits the mat and is rated in Hertz, or vibrations per minute.

The primary effect of vibratory frequency is its relationship to the compactor's working speed. Because the drum is moving into the mat, you need to make sure that these impacts are properly spaced. If the impact spacing is too wide, you can actually see impact marks at the surface of the mat. If the impact spacing is too narrow, you can see ridges in the surface of the mat. The correct impact spacing occurs when 26 to 46 impacts per meter (8 to 14 impacts per foot) are applied.

Many modern compactors have two vibratory frequencies, or sometimes frequencies that are variable. Frequencies are classified as low, medium or high.

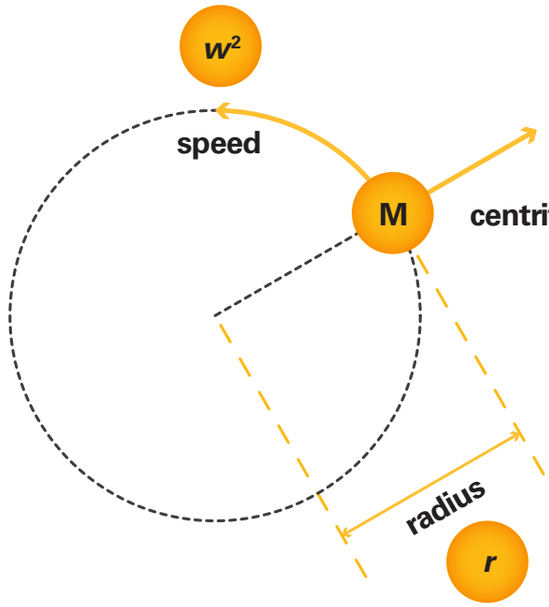
Low Frequency	40 to 47 Hz (2,400 to 2,800 vibrations per minute)
Medium Frequency	47 to 57 Hz (2,800 to 3,400 vibrations per minute)
High Frequency	Above 57 Hz (above 3,400 vibrations per minute)

[CONNECTING AMPLITUDE AND FREQUENCY]

Next, you must understand the relationship between amplitude and frequency. High amplitude is created when the eccentric weight is at its most off-center configuration. When the eccentric weight is in the most off-center or out-of-balance configuration, the eccentric weight shaft has to spin slowly in order to prevent excessive heat and wear on the weight shaft bearing. Therefore, high amplitude can only be associated with low vibratory frequency.

Low amplitude is created when the eccentric weight is in a more balanced configuration. When the eccentric weight shaft is more balanced, it can be spun more rapidly without damaging the vibratory drum components. Therefore, low amplitude can be associated with either high or low vibratory frequency.

On the job, the compaction team has to determine what are the correct vibratory characteristics in order to achieve density effectively and efficiently. If the mat requires a lot of force or energy in order to reach the specified density, then the crew will select a medium or high amplitude. When higher amplitudes are selected, a lower frequency will always be in use.



To calculate centrifugal force, the mass (M) of the eccentric weight is multiplied by the radius (r) of the eccentric weight rotation and by the speed of rotation (frequency) squared (w^2). The most significant factor in this equation is frequency.

[WHAT IS CENTRIFUGAL FORCE?]

Centrifugal force is a calculation that helps compactor designers establish the correct balance between drum weight, the mass of the eccentric weight and the speed of rotation of the eccentric weight. Centrifugal force has no practical meaning to the compactor operator or to quality control personnel.

There is often confusion about the meaning of centrifugal force as shown in compactor specification material. Many people think the higher the centrifugal force, the higher the compaction energy. This is an incorrect conclusion. A look at the formula for calculating centrifugal force will help clarify the situation.

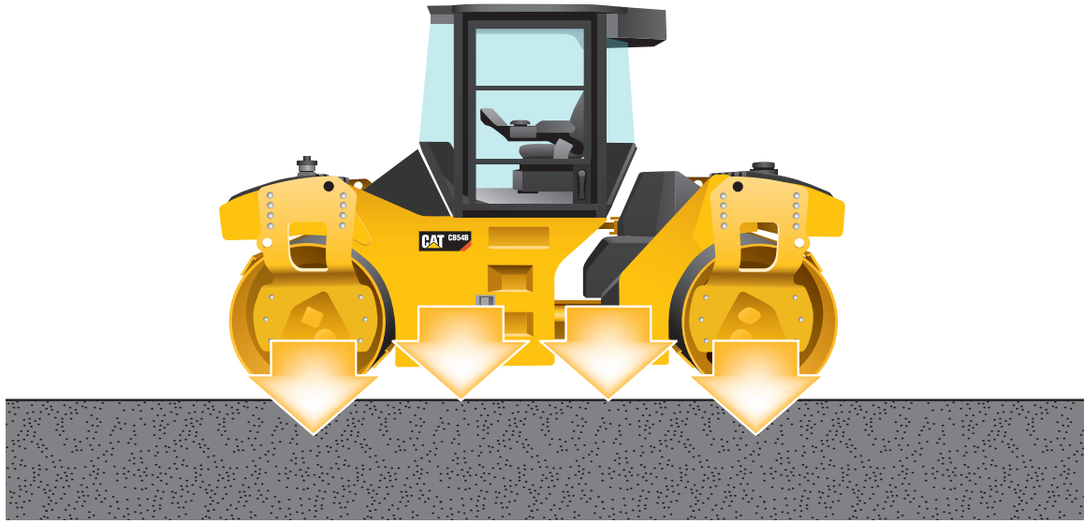
To calculate centrifugal force, the mass of the eccentric weight is multiplied by the radius of the eccentric weight rotation and by the speed of rotation (frequency) squared. The most significant factor in this equation is frequency. If frequency increases, centrifugal force is increased significantly. This concept is illustrated by looking at the vibratory system specifications for the Cat Tandem Drum Roller equipped with dual frequency and four amplitudes.

CAT VIBRATORY SYSTEM

Frequency	Amplitude	Centrifugal Force
42 Hz (2520 vpm)	High: 0.86 mm (0.034")	89 kN (19,980 lb)
42 Hz (2520 vpm)	Low: 0.73 mm (0.029")	75 kN (16,965 lb)
63 Hz (3800 vpm)	High: 0.44 mm (0.017")	103 kN (23,243 lb)
63 Hz (3800 vpm)	Low: 0.33 mm (0.013")	78 kN (17,438 lb)

On this machine, the highest centrifugal force numbers result when high frequency is selected. In high frequency, the amplitude, or impact force, is relatively small.

The lowest centrifugal force numbers result when low frequency is selected. As shown earlier, amplitudes are always higher when low frequency is selected. So, as the chart shows, higher centrifugal force does not necessarily correspond to higher compaction energy. Higher centrifugal force often means lower compaction energy. Compactor operators and quality control personnel are advised to ignore centrifugal force when considering vibratory system characteristics.

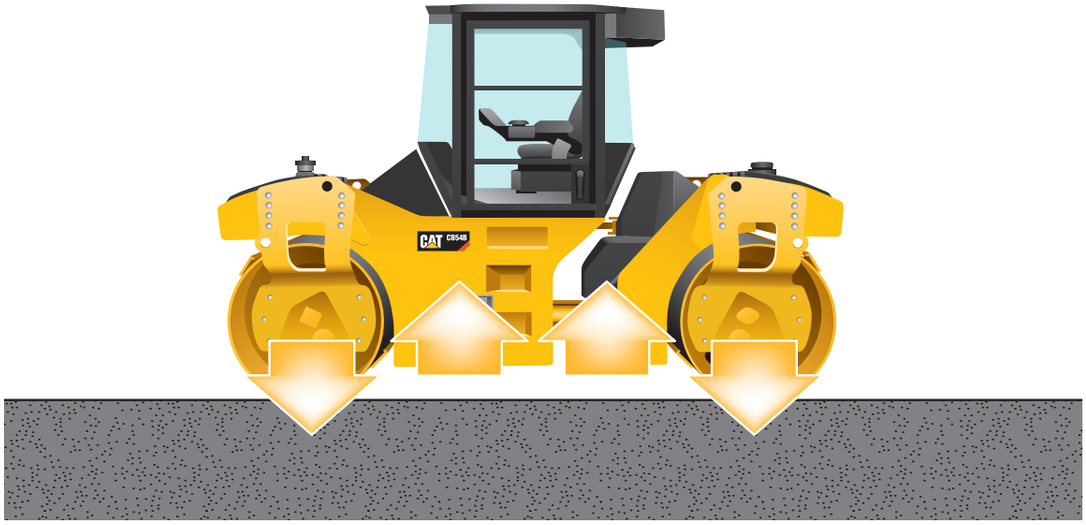


When compaction is balanced, most of the vibratory force is transmitted into the mat.

[BALANCED COMPACTOR VIBRATION]

When everything is in balance—amplitude, frequency, machine speed and drum weight—then impact and vibration forces are accepted by the mat. All vibratory characteristics are working together close to what is called system resonance and the machine operates smoothly. In this condition, most of the vibratory force is transmitted into the mat. The transmitted compaction force is maximized for the most productive operation.

Using the same machine on the same mat, you can select the same amplitude, but increase the frequency. Centrifugal force increases with the increase in frequency. The machine may no longer have the correct system resonance. In this condition, some of the compaction energy is not accepted by the mat, but instead is sent back up to the machine. The drums begin to lose contact with the mat. When the drums bounce, the operator loses steering control. Unbalanced vibration provides less effective compaction, may damage the mat, and is uncomfortable for the operator.



Some compaction energy will be transferred back to the machine if compaction is not balanced.

User Tip: *If drum bouncing occurs, try one of the following to restore smooth operation:*

- *Check the working speed to make sure you are operating in the range that produces 26 to 46 impacts per meter (8 to 14 impacts per foot).*
- *Switch to a lower amplitude setting.*
- *If available on the machine, switch to a higher frequency.*
- *Operate with one drum vibrating and one drum static.*
- *Operate in the static mode.*

Summary: *The forces of compaction and other machine characteristics are related. It is important for the crew to understand the relationship between the forces of compaction and how a mat accepts these forces. When the forces of compaction and other characteristics are aligned properly, efficient compaction will result.*


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Unit 3
FACTORS THAT AFFECT COMPACTION

Compaction is about more than experience—about more than what happened on the last job. Your compaction process will be more successful when you learn what information to gather, and how to interpret it.

In Unit Two, you learned about the forces of compaction and other factors that affect asphalt compaction.

Those factors, such as frequency, amplitude, working speed and drum width, can be controlled by the compaction crew.

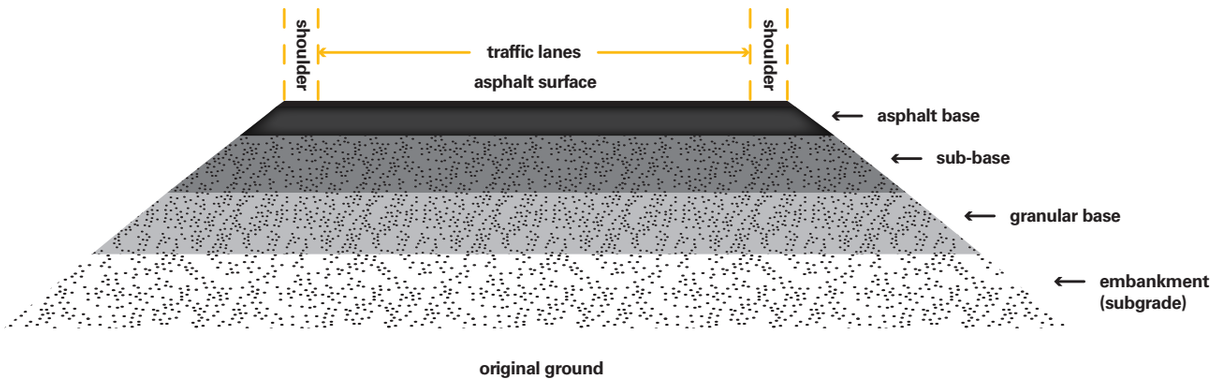
Many other factors that affect asphalt compaction cannot be controlled by compactor operators, on-site quality control personnel and supervisors.

Those factors include:

- Project design
- Mix design
- Asphalt layer thickness
- Mix temperature
- Climate conditions

It is important that operators and quality control personnel have information about these factors because they must consider them when developing the compaction techniques best suited for each unique project.

ROAD SECTION



[PROJECT DESIGN]

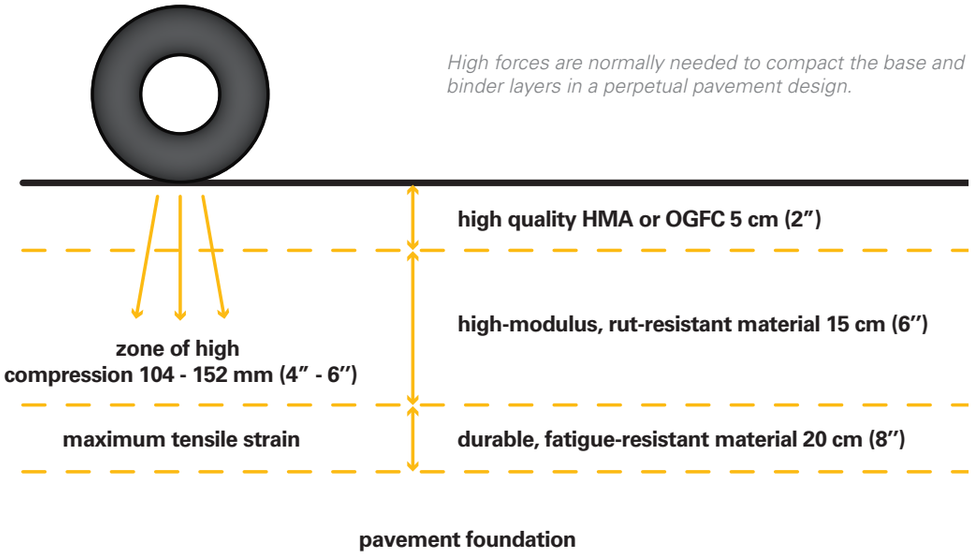
The design of the paving project affects compaction techniques. On new road construction or total reconstruction projects, multiple layers of asphalt will typically be applied. Each layer is made up of different material and has a different thickness. The first layer, called the base, is laid down on aggregate base or stabilized subgrade and normally consists of mix with fairly large aggregates. This is usually the thickest layer. It is being laid on a flexible base so high compaction energy is usually needed to achieve required density.

User Tip: *Any time you are compacting asphalt laid on a granular base, you need to remember that some of the compactor's impact force, or amplitude, will be absorbed by the somewhat flexible base. As you create your checklist for selecting amplitude, always consider the type of base under the asphalt layer. If it is a rigid base like a milled surface, you have to be careful not to use too much amplitude. If the base is flexible, there is less worry about using medium or high amplitudes.*

Often a second layer of asphalt, called the binder or intermediate layer, is used. The binder layer is usually thinner than the base layer and utilizes smaller aggregate. The binder layer normally requires less compaction energy to achieve the desired density. The base and binder layers contribute to structural support. The Intermediate course can contribute to drainage in conjunction with open-graded wearing courses.

Finally a surface layer, sometimes called the wear layer or friction layer, is applied. The surface layer is ordinarily the thinnest layer and is made up of the smallest aggregate. The surface layer is designed to be the stiffest layer and to contribute the most to pavement strength. Because the surface layer is usually relatively thin and laid down over a rigid surface, lower compaction energy is needed.

PERPETUAL PAVEMENT DESIGN



High forces are normally needed to compact the base and binder layers in a perpetual pavement design.

Perpetual pavement is a term used to describe another, long-life structural design. Perpetual pavement is designed to withstand an almost infinite number of axle loads without structural deterioration. Thick asphalt layers, sometimes totaling up to 60 cm (22"), limit the level of load-induced strain at the bottom of the asphalt layers.

Perpetual pavement design usually uses high-quality, deformation-resistant mixes. Most lower layers in the perpetual layer design require a high degree of compaction energy to reach the specified density.

Another very common pavement compaction application is compacting one or two fairly thin layers on a milled surface. Much of the work on asphalt structures now consists of maintenance through partial depth removal and repaving.

Following the cold planing step, there may be a leveling layer and then a surface layer. Seldom are these layers in excess of 50 mm (2"). Therefore, lower compaction energy is normally used for this application.



Compaction of thin layers on milled surfaces usually requires less compaction force.

[MIX DESIGN]

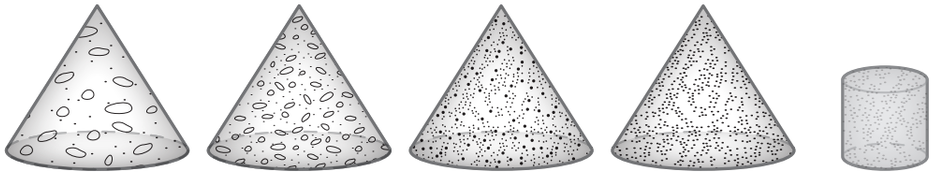
Asphalt is a generic term that includes many different types of mixtures of aggregates, fines, modifiers and asphalt cement produced in an asphalt plant at temperatures between 145° C and 190° C (300-350° F). A variation of traditional hot-mix asphalt is warm-mix asphalt. Warm-mix asphalt technology enables mix production and lay down at temperatures up to 40° C (100° F) lower than conventional hot-mix asphalt without sacrificing performance. In this manual, compaction of hot-mix asphalt and warm-mix asphalt designs is approached in the same manner.

Dense-graded mixes are produced with continuously graded aggregate. In other words, there are a variety of sizes of aggregates in the design. The design formula includes asphalt cement and fines. Typically the larger aggregates are surrounded by a matrix of mastic composed of asphalt cement and fines. The asphalt cement may be modified by materials such as polymer or latex rubber to develop additional stiffness.

Because the larger aggregates are surrounded by the mixture of asphalt cement and fines, there is less danger of damaging aggregates through the use of high compaction force. Depending on the layer thickness, medium to high amplitude is often selected when compacting dense-graded mixes.

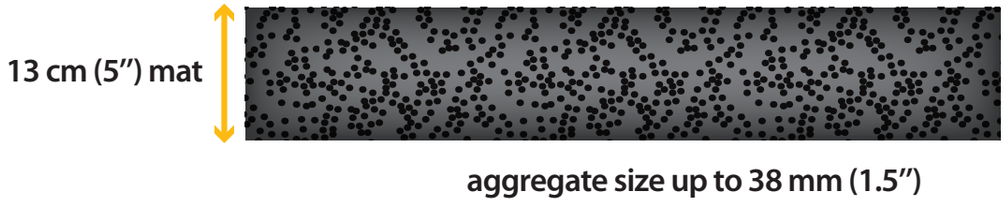
User Tip: *Continuing to add to the checklist for amplitude selection, the type of asphalt cement must be considered. If the asphalt cement has modifiers like polymers, fibers or latex rubber, the viscosity of the asphalt cement will be high. Moving the aggregates closer together during the compaction process will be more difficult due to the high viscosity of the modified asphalt cement. Therefore, you should be thinking of higher amplitudes when you know there is high-viscosity asphalt cement in the layer being compacted. Information about the type of asphalt cement can be found in the job mix formula that should be available to supervisors and quality control personnel.*

DENSE-GRADED MIX



Dense-graded mixes are often classified as either coarse or fine. Coarse mixes have a maximum aggregate size of 19 mm (3/4") or larger. Coarse mixes are usually laid down in fairly thick layers of 75 mm (3") or thicker. Mats that consist of coarse mix are less likely to move around under heavy compaction energy. You can use vibratory compactors in higher amplitude ranges and pneumatic compactors with higher ground pressures.

COARSE OR HARSH MIX



Dense-graded mixes with large aggregates are referred to as coarse mixes.

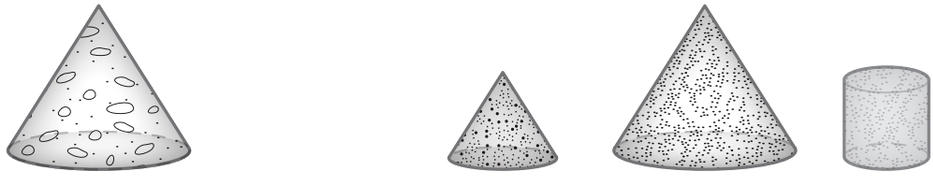
FINE OR TENDER MIX



Dense-graded mixes with smaller aggregates are referred to as fine mixes.

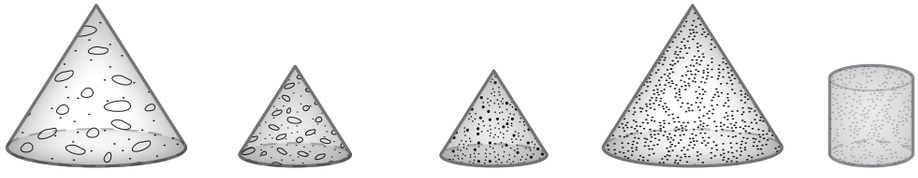
Some dense-graded mixes are classified as fine mixes. Fine mixes have a maximum aggregate up to 13 mm (1/2") and typically a fairly large percent of fines and asphalt cement. Some fine mixes can be unstable during the compaction process, especially if layer thickness exceeds 50 mm (2"). Static compaction may be needed to help stabilize fine mixes prior to vibratory passes. High compaction energy can damage layers of fine mix. Lighter compactors, either steel drum or pneumatic, are recommended for use on fine mixes.

OPEN-GRADED MIX



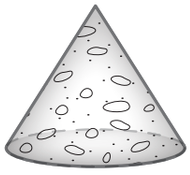
Open-graded mixes have relatively uniform-sized aggregate typified by an absence of intermediate sized particles. Mix designs typical of this structure are permeable friction layers and asphalt-treated permeable bases. Because of their open structures, precautions are taken to minimize the amount of asphalt cement draining to the bottom of the layer by using modified asphalt cement, usually latex rubber or fibers. Stone-on-stone contact with a heavy asphalt cement particle coating is typical for open-graded mixes.

GAP-GRADED MIX

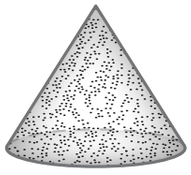


Gap-graded mixes use an aggregate gradation with particles ranging from large to fine with some intermediate sizes missing. Gap-graded mixes are also typified by stone-on-stone contact and are more permeable than dense-graded mixes.

STONE-MATRIX ASPHALT



nominal maximum aggregate



fines



asphalt cement

Stone-matrix asphalt (SMA), like other open-graded mixes, will be missing most intermediate aggregate sizes. However, stone-matrix asphalt mixes have a much higher proportion of fines. Modified asphalt cement combines with these fines to produce a thick mastic coating around and between the large aggregate particles.

User Tip: *Open-graded mixes, gap-graded mixes and stone-matrix asphalt all have more stone-on-stone contact than dense-graded mixes. Because of the high stone-on-stone contact, there is more likelihood that aggregates will be damaged during the compaction process. Lower amplitude settings on vibratory compactors or static compaction is usually recommended for these types of mixes. Method specifications that control the compaction process when these mixes are used may apply. Additionally, some SMA mixes that are heavily modified may be extremely stiff and may require more compaction force.*

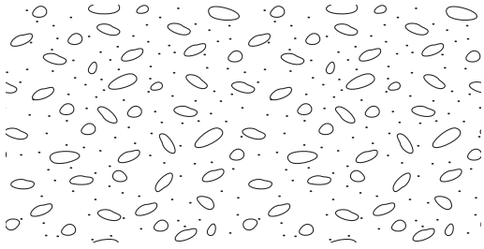
[AGGREGATE SHAPE]

Aggregate shape also affects compaction. The shape of the aggregate determines the amount of internal friction between the particles.

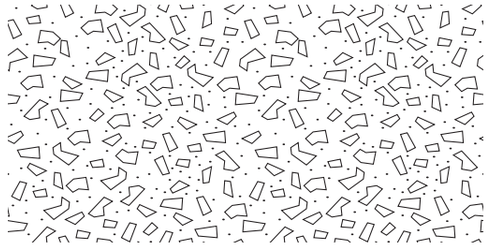
Rounded aggregates have low internal friction and move closer together within the layer under less compaction energy. However, mixes with rounded aggregates tend to be unstable and move around under the weight of the compactor. Therefore, when you know the shape of the aggregate is rounded, select low vibratory amplitude or a light static compactor.

Angular aggregates, on the other hand, have high internal friction. Once they are moved into contact with each other, angular aggregates produce high pavement strength. More force and heavier compactors are needed to overcome the internal friction between the fractured faces of crushed stone aggregate. Most high-traffic mix designs specify the use of crushed aggregates with a certain number and shape of fractured faces.

ROUNDED AGGREGATES
low internal friction



ANGULAR AGGREGATES
high internal friction



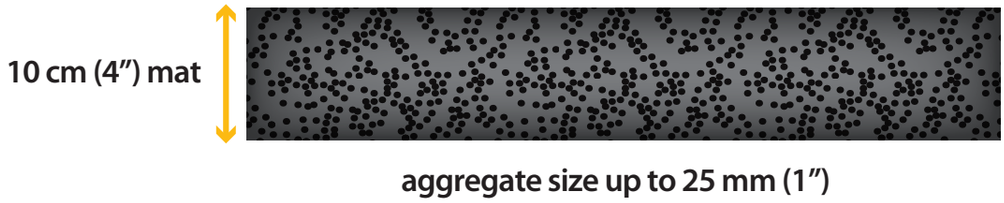
[LAYER THICKNESS / AGGREGATE SIZE]

No matter what type of mix or what type of project, a very important factor is the ratio between the size of the largest aggregate in the mix and the thickness of the layer. This ratio strongly affects the ability of the mat to accept compaction energy and reach the specified density.

For example, a layer thickness of 10 cm (4") and a maximum aggregate size of 25 mm (1") is relatively easy to compact. You can use high compaction forces and not worry about damaging aggregates.

With a 4:1 ratio, there is ample space for aggregate movement allowing the aggregates to re-orient. Many public works departments specify a 3:1 ratio as a minimum for high-traffic mix design.

4:1 RATIO LAYER THICKNESS TO AGGREGATE SIZE



2:1 RATIO LAYER THICKNESS TO AGGREGATE SIZE



When the ratio of layer thickness to aggregate size is less than 3:1, the compaction process is much more difficult. In particular, when there are angular aggregates in the layer, it is likely that those aggregates will not move into the proper compacted orientation without being damaged. The compactor operator is more likely to notice drum bouncing or the appearance of uncoated rock at the surface of the mat. Lack of density and damaged aggregates will lead to premature mat failure. When the ratio of layer thickness to aggregate size is less than 3:1, low compaction energy is needed.



Mat thickness can be variable when slope control is used by the paving crew.

User Tip: *Even though the design may call for the required minimum 3:1 ratio, there are instances when the ratio of layer thickness to aggregate size will be less. The most common situation occurs when slope control is used on the paving equipment to create profile. Slope control of the paver screed means that mat thickness will probably be variable across the width of the mat. The left half of the mat may be 75 mm to 50 mm (3-2") thick and the right half of the mat may be 50 mm to 25 mm (2-1") thick. The left side of the mat will compact normally, but the drums will probably begin to bounce when the compactor moves over to the right side. Less compaction energy is needed on the right side. The easiest solution to the problem is operating the compactor with one drum vibrating and one drum static, or both drums static. Changing vibratory system operation can be done easily with the flip of a switch at the operator station. The operator does not have to get off the machine and change the amplitude setting to overcome the drum bouncing.*

[MIX TEMPERATURE]

The next factor, temperature of the asphalt layer, has a major effect on compaction. Creating density in any asphalt layer normally is the easiest at the highest possible temperature. At high temperature, the asphalt cement that is part of the mix is at its lowest viscosity. The aggregates in the mix move closer together easily when the asphalt cement is fluid, or at its lowest viscosity. The asphalt cement becomes stiffer as it cools. The aggregates in the mix lock in position and no more air can be forced out.

Depending on how a mix performs at high temperatures, the upper limit that permits compaction is approximately 160° C (320° F).

Some types of mixes may be unstable at high temperatures and will move in front of the drum rather than consolidate under the drum. Mixes that are unstable at high temperatures usually have a high percentage of small aggregate, fines and asphalt cement. When a mix deforms due to high temperatures, the solution is to stay farther behind the paver to allow the mat to cool enough to permit normal compaction.



Most, if not all, the required layer density should be completed by the time the mat cools to about 90° C (190° F). At this temperature the asphalt becomes so stiff that no additional aggregate movement is possible. While you may be able to clean up marks in the surface of the mat, it's not likely that you will get additional density. If the mat is cooling before required density is reached:

- Work closer to the paver
- Increase compaction energy
- Add more compactors

There are exceptions to the general rules concerning upper and lower temperature limits for compaction. Some mixes have a tender zone that is between the traditional upper and lower temperature limits. The mat will compact normally in a very high temperature range. Then it will become unstable in an intermediate temperature range, usually called the tender zone.



One sign that the compactor is working in a tender zone is a rolled over and cracked drum edge cut mark.

If a compactor operates on the mat during the tender zone, you will normally see the mat moving in front of the drum or pneumatic tires. Another sign that you are operating in the tender zone will be the appearance of a rolled-over and cracked drum edge cut mark. Normally the drum edge leaves a straight cut mark. The rolled-over cut mark that indicates the compactor is in the tender zone is probably easier for the operator to see.

When compacting a mix that has a tender zone, you need to achieve as much density as possible in the high temperature zone. Compaction should normally be stopped when the mat enters the tender zone. When the tender zone ends, the mat normally becomes stable again and additional density can be achieved. The mat should be close to final density before the tender zone starts. When the tender zone ends and compaction starts again, select low amplitude to avoid drum bouncing on the cool, dense mat.

[CLIMATE CONDITIONS]

The final factor that influences compaction is climate, primarily ambient temperature and wind conditions. On hot, sunny days with high ambient temperatures, the mat retains its heat longer, so the period of workability is increased.

On cool and windy days, the mat loses heat faster. A crust can actually form over the surface of the mix and prevent the compaction forces from uniformly penetrating into the mat.

Checking mat temperature is done in two ways. The most common is to use an infrared temperature scanner. The scanner provides a quick check of the surface temperature at a certain point. The operator or quality control technician can quickly monitor mat temperatures at various locations. The other way to check mat temperature is using a probe thermometer. The probe shows the internal temperature of the mat and is a better indicator of how the mix will react to the forces of compaction.



User Tip: *Compactor operators may have to adjust their rolling patterns as ambient conditions change throughout the day. In the morning when the temperature is normally the lowest, a short rolling pattern may be used in order to stay very close to the paver. As the ambient temperature increases and the mat stays hot longer, the compactor operators can increase the length of their rolling patterns and not worry as much about staying close to the paver.*

All these temperature and mix design factors are important because they determine how much time you have to get the required density before the mat cools to below 90° C (190° F). You can also use this information to determine how much time you have before the start of a tender zone, if any exists.

In the past, quality control personnel had to use engineering charts to determine the time available for compaction based on layer thickness, layer temperature behind the paver and ambient conditions. Today, there are several software programs available to calculate the time with more accuracy and faster. One type of software is called PaveCool (PaveCoolVersion 2.400 copyright 2001-2005 Minnesota Department of Transportation). A series of PaveCool calculations follows.

User Tip: *Mat surface temperature is always lower than the mat's internal temperature. If available, use a probe to check internal temperature. Then use the infrared scanner to check surface temperature at the same location. You may find that the surface temperature is up to 15° C (30° F) cooler than the internal temperature. Knowing the temperature differential will allow you to correlate the surface temperature shown by the scanner to the actual internal temperature.*

User Tip: *Caterpillar recommends that quality control personnel develop a series of cooling curves prior to the start of a project. The cooling curves should represent any changes in layer thickness and should anticipate changes in mix temperature and ambient conditions.*

User Tip: *Cooling curves are also extremely useful when compacting mixes that have a tender zone. You can set the PaveCool software to show the length of time prior to the start of tenderness and the length of time that tenderness will persist.*

[COOLING CURVES]

Start Date/Time [17/6/2011 ▾] [11:52 AM]

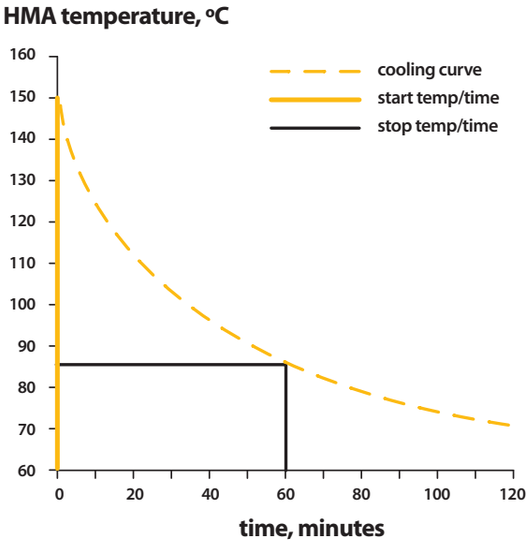
Environmental Conditions
Air Temperature: [16] °C
Wind Speed: [8] km/h
Sky: [Clear & Dry ▾]
Latitude: [45] °N

Mix Specifications
Mix Type: [Fine/Dense Graded ▾]
Binder Grade, PG: [58 ▾] [34 ▾]
Lift Thickness: [75] mm
Delivery Temperature: [150] °C

Existing Surface
Material Type: [Granular Base ▾]
Material Condition: [Dry ▾] [Unfrozen ▾]
Surface Temperature: [16] °C

Recommended Times
Start Rolling: [0] minutes after laydown
Stop Rolling: [61] minutes after laydown

Units ◉ SI ○ English



PaveCool software requires input on environmental conditions, mix specifications, and the existing surface. In this example, the air temperature is 16° C (60° F), the wind is light, it is sunny and the project is located near Paris, France (45° north latitude). The asphalt layer consists of dense-graded material laid 75 mm (3") thick, and passing under the paver screed at 150° C (302° F). The asphalt layer is being laid down on granular material that is also 16° C (61° F). The resulting cooling curve shows that there are 61 minutes for compaction before the mat cools to the minimum temperature that has been set at 85° C (185° F).

Start Date/Time [17/6/2011 ▾] [11:52 AM]

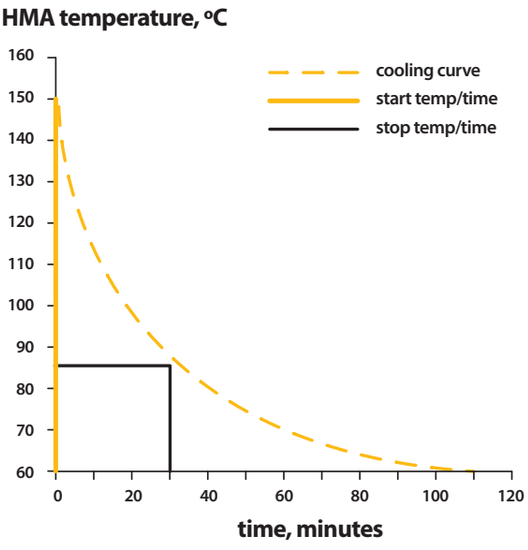
Environmental Conditions
Air Temperature: [16] °C
Wind Speed: [8] km/h
Sky: [Clear & Dry ▾]
Latitude: [45] °N

Mix Specifications
Mix Type: [Fine/Dense Graded ▾]
Binder Grade, PG: [58 ▾] [34 ▾]
Lift Thickness: [50] mm
Delivery Temperature: [150] °C

Existing Surface
Material Type: [Granular Base ▾]
Material Condition: [Dry ▾] [Unfrozen ▾]
Surface Temperature: [16] °C

Recommended Times
Start Rolling: [0] minutes after laydown
Stop Rolling: [31] minutes after laydown

Units ◉ SI ○ English



In this example, all the inputs are the same except the layer thickness has been changed to 50 mm (2"). When the new cooling curve is calculated, the time available for compaction had decreased by 50%. Time available for compaction is now 31 minutes.

Start Date/Time [17/6/2011 ▾] [11:52 AM]

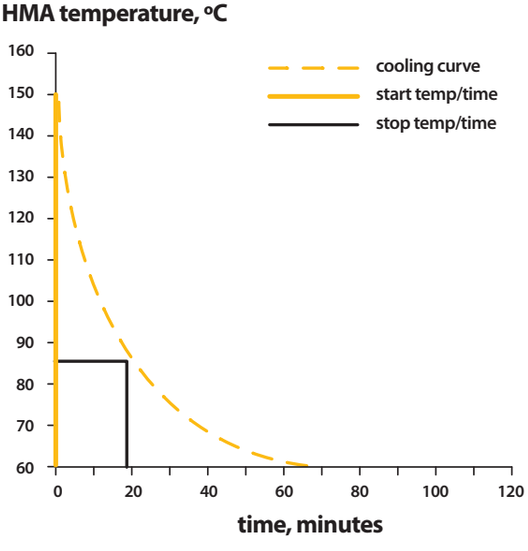
Environmental Conditions
Air Temperature: [16] °C
Wind Speed: [8] km/h
Sky: [Clear & Dry ▾]
Latitude: [45] °N

Mix Specifications
Mix Type: [Fine/Dense Graded ▾]
Binder Grade, PG: [58 ▾] [34 ▾]
Lift Thickness: [38] mm
Delivery Temperature: [150] °C

Existing Surface
Material Type: [Granular Base ▾]
Material Condition: [Dry ▾] [Unfrozen ▾]
Surface Temperature: [16] °C

Recommended Times
Start Rolling: [0] minutes after laydown
Stop Rolling: [19] minutes after laydown

Units ◉ SI ○ English



Next, assume that you are laying down a relatively thin surface layer that is 38 mm (1.5") thick. Everything else remains the same. The cooling curve shows that only 19 minutes are available for compaction.

Layer thickness has a major impact on the time available for compaction. The thinner the layer, the more difficult the compaction process. Knowing how much time is available to get density on thin mats will help you determine rolling patterns and the number of compactors needed for the project.



Offset drum capability facilitates wider coverage for faster compaction on thin mats that lose heat rapidly.

User Tip: *Some compactors feature the ability to offset the front and rear drums to almost double the width of the compaction coverage. Therefore, fewer overlapping passes are generally required to cover the width of the mat. With the drums offset, the amount of compaction force delivered to the mat is reduced. However, thin mats require less compaction force and the compactor is able to work closer to the paver where the mat is hotter and more susceptible to compaction.*

Start Date/Time [17/6/2011 ▾] [11:52 AM]

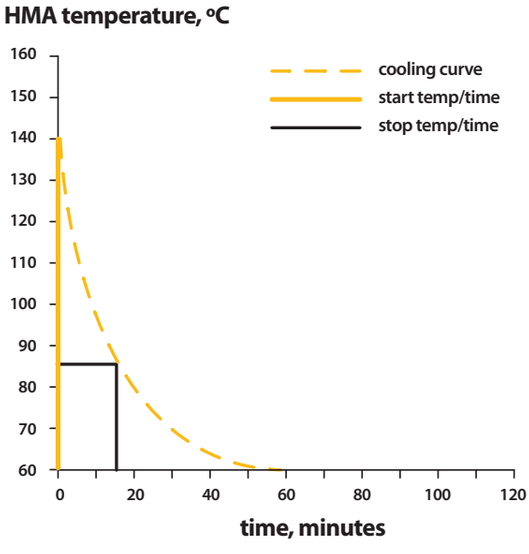
Environmental Conditions
Air Temperature: [16] °C
Wind Speed: [8] km/h
Sky: [Clear & Dry ▾]
Latitude: [45] °N

Mix Specifications
Mix Type: [Fine/Dense Graded ▾]
Binder Grade, PG: [58 ▾] [34 ▾]
Lift Thickness: [38] mm
Delivery Temperature: [140] °C

Existing Surface
Material Type: [Granular Base ▾]
Material Condition: [Dry ▾] [Unfrozen ▾]
Surface Temperature: [16] °C

Recommended Times
Start Rolling: [0] minutes after laydown
Stop Rolling: [16] minutes after laydown

Units ◉ SI ○ English



Next, consider how mat temperature affects the time available for compaction. In this example, mat thickness is 38 mm (1.5"), but the temperature of the mat passing under the screed is lowered to 140° C (284° F). Compared to mat temperature of 150° C (302° F) where 19 minutes was available for compaction, the slightly cooler mat reduces the time available for compaction to 16 minutes.

Start Date/Time [17/6/2011 ▾] [11:52 AM]

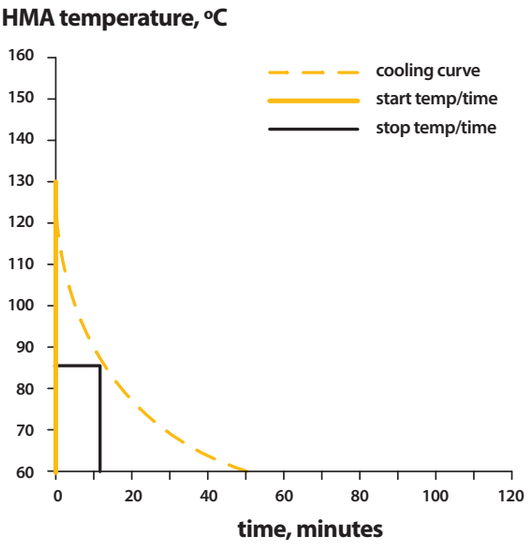
Environmental Conditions
Air Temperature: [16] °C
Wind Speed: [8] km/h
Sky: [Clear & Dry ▾]
Latitude: [45] °N

Mix Specifications
Mix Type: [Fine/Dense Graded ▾]
Binder Grade, PG: [58 ▾] [34 ▾]
Lift Thickness: [38] mm
Delivery Temperature: [130] °C

Existing Surface
Material Type: [Granular Base ▾]
Material Condition: [Dry ▾] [Unfrozen ▾]
Surface Temperature: [16] °C

Recommended Times
Start Rolling: [0] minutes after laydown
Stop Rolling: [12] minutes after laydown

Units ◉ SI ○ English



Finally, when the mat temperature is lowered to 130° C (266° F), the time available for compaction drops to 12 minutes. Lowering the mat temperature does reduce the time available for compaction but not to the extent that mat thickness affects time available for compaction.

Start Date/Time [17/6/2011 ▾] [11:52 AM]

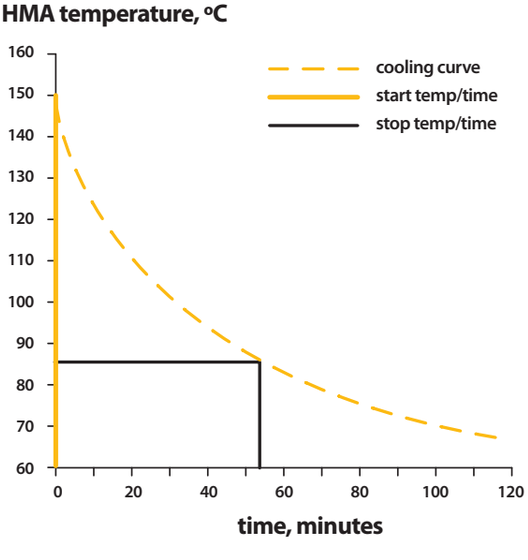
Environmental Conditions
Air Temperature: [10] °C
Wind Speed: [8] km/h
Sky: [Clear & Dry ▾]
Latitude: [45] °N

Mix Specifications
Mix Type: [Fine/Dense Graded ▾]
Binder Grade, PG: [58 ▾] [34 ▾]
Lift Thickness: [75] mm
Delivery Temperature: [150] °C

Existing Surface
Material Type: [Granular Base ▾]
Material Condition: [Dry ▾] [Unfrozen ▾]
Surface Temperature: [10] °C

Recommended Times
Start Rolling: [0] minutes after laydown
Stop Rolling: [54] minutes after laydown

Units ◉ SI ○ English



Now consider the affect of ambient temperature on the time available for compaction. In this example, mat thickness reverts to 75 mm (3"). When the ambient temperature was 16° C (61° F), there were 61 minutes available for compaction. When the ambient temperature is reduced to 10° C (50° F), the time available for compaction is reduced by about 10% down to 54 minutes.

Start Date/Time [17/6/2011 ▾] [11:52 AM]

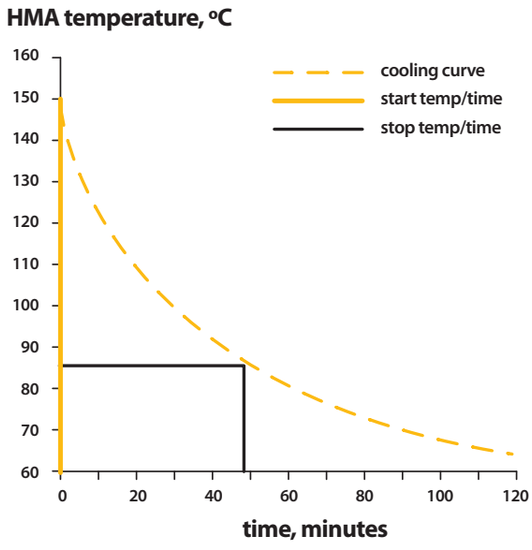
Environmental Conditions
Air Temperature: [5] °C
Wind Speed: [8] km/h
Sky: [Clear & Dry ▾]
Latitude: [45] °N

Mix Specifications
Mix Type: [Fine/Dense Graded ▾]
Binder Grade, PG: [58 ▾] [34 ▾]
Lift Thickness: [75] mm
Delivery Temperature: [150] °C

Existing Surface
Material Type: [Granular Base ▾]
Material Condition: [Dry ▾] [Unfrozen ▾]
Surface Temperature: [5] °C

Recommended Times
Start Rolling: [0] minutes after laydown
Stop Rolling: [49] minutes after laydown

Units ◉ SI ○ English



Finally, the ambient temperature is reduced to 5° C (41° F). Time available for compaction is reduced again to 49 minutes. It is evident that reducing ambient temperature reduces time available for compaction, however, not to the extent resulting from reduced mat thickness.

Start Date/Time [17/6/2011 ▾] [11:52 AM]

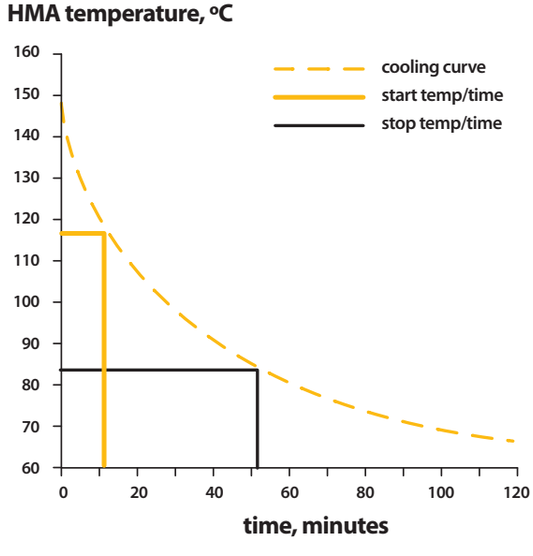
Environmental Conditions
Air Temperature: [16] °C
Wind Speed: [8] km/h
Sky: [Clear & Dry ▾]
Latitude: [45] °N

Mix Specifications
Mix Type: [Fine/Dense Graded ▾]
Binder Grade, PG: [58 ▾] [34 ▾]
Lift Thickness: [65] mm
Delivery Temperature: [150] °C

Existing Surface
Material Type: [Granular Base ▾]
Material Condition: [Dry ▾] [Unfrozen ▾]
Surface Temperature: [16] °C

Recommended Times
Start Rolling: [12] minutes after laydown
Stop Rolling: [52] minutes after laydown

Units ◉ SI ○ English



In this example, the ambient temperature is 16° C (61° F). We are laying down a dense-graded mix at a depth of 65 mm (2.5"). The mix is passing under the screed at 150° C (302° F). Documentation indicates that the mix begins to display tenderness at 115° C (239° F) and that the mat becomes stable again at 85° C (185° F). The cooling curve shows that there are 12 minutes available for compaction prior to the start of the tender zone. You will want to achieve as much density as possible during this 12-minute period and then stay off the mat for 40 minutes. After the tender zone disappears, high frequency with low amplitude is commonly used to gain additional density if needed. Or, static rolling to clean up any drum marks may be all that is necessary. Again, knowing the time available for compaction before the tender zone will help you set up the rolling pattern and determine the number and types of compaction equipment.